

Admin User Guide

Cebuana Lhuillier Parking Reservation System

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1. Overview

As an Admin, you have full control over the Parking Reservation System. Your responsibilities include:

- Managing users (create, edit, block, bulk upload, tag)
- Managing buildings, floors, and parking slots (with floor layout images)
- Configuring zones, parking rules, building policies, and event blocks
- Viewing reports, generating AI insights, exporting reservations
- Managing attendants, site branding, and (where enabled) the active database backend

What's new in v2.4 — comprehensive **Policy Capabilities Matrix** (§6), runtime feature flags including AI Insights enable/disable (§10.4) and **Attendant Mode / Self Check-In** (§6.3), database switcher (§14), and floor-layout images now persist on object storage in the production cluster.

2. Accessing the Admin Panel

1. Log in with your admin credentials.
2. You will be redirected to the **Admin Dashboard** at `/admin`.
3. Use the **sidebar navigation** to access all admin modules.

Admin Navigation

Menu Item	Page	Description
Dashboard	<code>/admin</code>	Overview stats and charts
Users	<code>/admin/users</code>	User management with bulk actions
Buildings	<code>/admin/buildings</code>	Building, floor, and slot management
Zones	<code>/admin/zones</code>	Zone grouping and user assignment

Menu Item	Page	Description
Parking Config	/admin/parking-config	Per-building booking rules and exclusivity
Attendants	/admin/attendants	Attendant management
Reservations	/admin/reservations	View/cancel all reservations (with reason)
Event Blocking	/admin/event-blocking	Block parking slots for events
Reports	/admin/reports	Analytics and AI insights
Settings	/admin/settings	Login page customization

3. User Management

3.1 Viewing Users

Navigate to **Users** in the sidebar. The user table shows:

- Name, email, company, role
- **Job Family** (e.g., Department Head)
- Main building, tags (VIP/Group Head)
- Status (Active/Blocked)
- Created and updated timestamps

3.2 Filtering Users

Use the filter bar above the table:

- **Search** — Type to filter by name, email, or company
- **Role** — Filter by User, Admin, or Attendant
- **Company** — Filter by company name
- **Zone** — Filter by zone assignment (includes "No Zone" option)
- **Main Building** — Filter by building (includes "No Building" option)

The result count updates dynamically as you apply filters.

3.3 Creating a Single User

1. Click **"Add User"**.
2. Fill in the form:
3. Email (required, must be unique)
4. First Name, Last Name (required)
5. Company
6. Role (User, Admin, or Attendant)

7. Password
8. **Job Family** (optional — e.g., "Department Head", "Manager", "Staff")
9. Main Building (optional)
10. Assigned Buildings (for attendants)
11. Default Start/End Time (optional)
12. Click **Save**.

The user will be created with `must_change_password: true` and must change their password on first login.

3.4 Bulk Uploading Users

1. Click **"Bulk Upload"** in the User Management page.
2. Optionally click **"Download Template"** to get a CSV template with the correct columns.
3. Prepare your CSV with columns:
4. `email` (required)
5. `first_name` (required)
6. `last_name` (required)
7. `company` (optional)
8. `role` (optional, defaults to "user")
9. `job_family` (optional — e.g., "Department Head")
10. `main_building` (optional — use the **building name**, the system resolves it to the building ID)
11. `zone` (optional — use the **zone name**, the system automatically assigns users to the zone)
12. Set a **default password** for all new users.
13. Select the file and click **Upload**.
14. Review the upload summary showing created, skipped (duplicate emails), zone assignments, and any errors.

3.5 Bulk Actions (Multi-Select)

1. Use the **checkboxes** in the user table to select multiple users.
2. Or click **"Select all X filtered"** to select all users matching the current filter.
3. The **bulk action bar** appears with options:
4. **Set Main Building** — Assign a main building to all selected users
5. **Assign to Zone** — Add all selected users to a specific zone

3.6 Individual User Actions

Click the **actions dropdown** (three dots) on any user row to:

- **Edit** — Modify user details
- **Reset Password** — Set a new temporary password (user must change on next login)
- **Block / Unblock** — Prevent or restore user access
- **Set Tags** — Assign VIP or Group Head tags
- **Delete** — Permanently remove the user

3.7 Tags

Tags grant special privileges:

- **VIP** — Can book multi-day reservations in non-main buildings (within zone) without the single-day restriction
 - **Group Head** — Same privileges as VIP
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4. Building Management

4.1 Creating a Building

1. Navigate to **Buildings**.
2. Click "**Add Building**".
3. Enter:
4. Building name
5. Address
6. Number of floors
7. Slots per floor (auto-generated with prefix)
8. Slot prefix (e.g., **F1-** for Floor 1 → slots **F1-1** , **F1-2** , etc.)
9. Click **Create**.

Floors and slots are automatically created.

4.2 Managing Floors

Each building card shows its floors. For each floor you can:

- **Add Floor** — Create a new floor with custom slot labels or auto-generated slots
- **Upload Layout** — Upload a floor plan image (PNG, JPG, JPEG, GIF, WebP; max 10MB)
- **Delete Layout** — Remove the floor plan image

4.3 Managing Slots

Within each floor, you can see all parking slots. For each slot:

- **Rename** — Change the slot label
- **Block (Maintenance)** — Mark as under maintenance; the slot becomes unavailable for booking
- **Unblock** — Restore the slot to active status
- **Delete** — Permanently remove the slot (only if no active reservations)

4.4 Adding Slots to an Existing Floor

1. Click **"Add Slots"** on the floor card.
2. Either:
3. Enter **custom labels** (comma-separated, e.g., `VIP-1, VIP-2, VIP-3`)
4. Or specify a **count** with an auto-prefix
5. Click **Add**.

4.5 Deleting a Building

Click **Delete** on the building card. This will **cascade delete** all floors and slots within the building.

Warning: This operation is irreversible. Ensure no active reservations exist for the building's slots.

5. Zone Management

Zones group buildings together and control which users can book in which buildings.

5.1 Creating a Zone

1. Navigate to **Zones**.
2. Click **"Create Zone"**.
3. Enter a zone name (e.g., "Metro Manila Zone").
4. Select the **buildings** to include in the zone.
5. Select the **users** to assign to the zone.
6. Click **Create**.

5.2 Editing a Zone

Click **Edit** on a zone card to:

- Rename the zone
- Add or remove buildings
- Add or remove users

5.3 Zone Booking Rules

- Users assigned to a zone can book in **any building** within that zone.
- Booking in a building that is **not** the user's main building requires:
 - **Single-day** booking only
 - A **reason** for the visit
- VIP/Group Head users are exempt from the single-day restriction.
- Users **cannot** book in buildings outside their assigned zones.

5.4 Deleting a Zone

Click **Delete** on the zone card. This removes the zone but does not delete the buildings or users.

6. Policy Capabilities Matrix

The platform exposes **five layers of configurable policy**. Use this matrix to find the right place to enforce a rule before reaching for a workaround. Every cell links to the section that explains *how* to configure it.

Layer	What it controls	Where to configure	Highlights
System-wide	Site branding, login content, optional features (AI Insights, Attendant Mode), active database	Settings (§11) · System badge (§14) · Feature flags (§6.3)	Tri-state feature flag for AI Insights · Attendant Mode toggle with configurable self-check-in window · live login-page content editor · multi-DB support (MongoDB / Couchbase Capella / Couchbase Enterprise)
Zone	Which users can book in which buildings; main-building rules	Zones (§5)	Group buildings into zones · zone-restricted bookings · "main building" assignment per user · single-day / reason-required cross-zone bookings
Building (operational)	Per-building booking knobs that	Parking Config (§7)	Booking window, release time, default times, no-show auto-release, waitlist

Layer	What it controls	Where to configure	Highlights
	change behaviour daily		on/off, main-building exclusivity , notification windows
Building (policy)	Per-building access rules tied to vehicles & slots	Building Policies (§12)	Sticker enforcement · slot-level vehicle binding · open floors · per-slot user assignment · zone-filtered user picker
Time-bound override	Reserve N slots for a specific event window	Event Blocking (§13)	Block any combination of slots on a floor for a date/time range · stamped with name + reason · auto-frees on delete

6.1 Quick reference — "I want to..."

I want to...	Use this layer	Specific setting
Stop people from booking 30 days ahead	Building (operational)	<code>booking_window_days</code> — caps at 1–30, default 7
Free a slot automatically when someone doesn't show up	Building (operational)	<code>no_show_release_enabled</code> + <code>no_show_release_minutes</code>
Let users queue up when a building is full	Building (operational)	<code>waitlist_enabled</code> + <code>waitlist_notification_window_minutes</code>
Restrict a building to "home base" employees only	Building (operational)	<code>main_building_exclusive: true</code>
Tie a parking sticker to one specific vehicle	Building (policy)	<code>requires_sticker: true</code> + slot registration with vehicle plate
Let one designated user own slot B1-03 with vehicle ABC1234	Building (policy)	Slot registration (user + vehicle plate per slot)
Make floor 4 a free-for-all (no slot ownership)	Building (policy)	Add floor to <code>open_floor_ids</code>
Reserve 12 slots for the board meeting next Friday	Time-bound override	Event Blocking → pick slots → name + reason
Let only Manila employees book Manila buildings	Zone	Create Zone "Manila", assign buildings + users

I want to...	Use this layer	Specific setting
Hide AI Insights from the admin UI in air-gapped deployments	System-wide	<code>AI_INSIGHTS_ENABLED=false</code> env var (§9.4)

6.2 Validation order (so you can predict what happens)

Whenever a user tries to book, the system checks rules **in this order** and stops at the first failure with a clear error message:

1. **Auth & block status** — user is logged in and not blocked.
2. **Vehicle ownership** — the chosen vehicle belongs to the user; plate is globally unique.
3. **Time guards** — date is within `booking_window_days`; not in the past; not overlapping another reservation; not before `release_time`.
4. **Zone access** — the building lives in a zone the user is in.
5. **Main-building exclusivity** — if the building has `main_building_exclusive: true`, the user's `main_building` must equal this building (admins exempt).
6. **Cross-zone single-day rule** — booking in a non-main building → date range must be a single day, reason required (VIPs exempt).
7. **Building policy** — sticker required? Vehicle has a slot registration in this building? On a non-open floor, is the vehicle bound to *this specific slot*?
8. **Event block** — slot isn't already locked by an event block for the requested time window.
9. **Slot availability** — slot is `available` and not currently held by another reservation.

Knowing the order helps you give users accurate guidance: "you got error X, so checks 1–N already passed."

6.3 Attendant Mode vs. Self Check-In

The platform supports **two arrival-confirmation modes**, set via a single environment variable on the backend (`ATTENDANT_MODE_ENABLED`). The mode is read live by both backend and frontend through `GET /api/system/features`, so changes take effect on the next page load — no rebuild.

Mode	Toggle	What happens when the parker arrives
Attendant Mode (default)	<code>ATTENDANT_MODE_ENABLED=true</code> (or unset)	On-site attendants log into the Attendant Dashboard, see today's reservations, scan/confirm arrivals, or report no-shows. The auto-no-show loop sweeps any reservation whose <code>end_time</code> has passed without confirmation.

Mode	Toggle	What happens when the parker arrives
Self Check-In	<code>ATTENDANT_MODE_ENABLED=false</code>	Attendants are not used at this site. The parker sees a "Check in" button on their reservation card that becomes active at <code>start_time</code> and stays active for <code>SELF_CHECKIN_WINDOW_MINUTES</code> (default 15 , configurable 5–240). Tapping it confirms the slot. If the window elapses with no tap, a background loop marks the reservation as no-show, releases the slot immediately , and notifies the next person on the waitlist.

Where the check-in button appears

- User → **My Reservations** page → today's PENDING reservation card → green **"Check in"** button on the right of the row.
- Visible only when (a) the system is in Self Check-In mode, (b) the reservation is the parker's own, (c) current time is between `start_time` and `start_time + SELF_CHECKIN_WINDOW_MINUTES`.

Error messages your parkers will see

Situation	HTTP	Message
Too early — <code>start_time</code> hasn't arrived	400	"Too early to check in. Your reservation starts in ~N minutes."
Inside the window, success	200	"Checked in — your slot is confirmed."
Tapped twice (idempotent)	200	"Already checked in"
Window has expired	410	"Check-in window has expired (N min after start). The slot will be released shortly."
Mode is currently in attendant mode	403	"Self check-in is disabled. An attendant will confirm your arrival."

Choosing the window length

- 15 min (default) — typical for office buildings; gives drivers a buffer for traffic / finding the entrance.
- 5 min — high-demand sites where you want the slot to recycle fast.
- 30–60 min — sites with long approach roads or strict gate procedures.

No-show behaviour

Whichever mode you're in, a no-show:

1. Sets reservation status to `no_show`.

2. Increments the user's `no_show_count` (visible to admins on the Users page).
3. Sends an in-app notification to the parker.
4. Frees the slot — **immediately** in Self Check-In mode, or after `no_show_release_minutes` if attendant mode is on and the building has `no_show_release_enabled`.
5. Calls the waitlist notifier — the next person on the waitlist for that building/date gets an in-app notification that a slot is now available.

Operational tip: switching from attendant mode to self check-in is a configuration change, not a feature deployment — you can A/B different buildings by spinning up two backend pods with different env vars and routing traffic accordingly. Most customers won't need that; one flag for the whole estate is fine.

7. Parking Configuration

Configure booking policies per building.

7.1 Editing Configuration

1. Navigate to **Parking Config**.
2. Select a **Building**.
3. Configure:

Setting	Description	Default
Default Start Time	Pre-filled start time for bookings	08:00
Default End Time	Pre-filled end time for bookings	18:00
Release Time	Time when slots become bookable each day	06:00
Booking Window (days)	How many days in advance users can book	7
Auto No-Show Release	Automatically release slots from no-show reservations	Off
No-Show Release Minutes	Minutes after being tagged as no-show before slot is released	30
Waitlist Enabled	Allow users to join a waitlist when all slots are booked	Off
Waitlist Notification Window (min)	Minutes a notified waitlist user has to book before the slot is offered to the next person	15

Setting	Description	Default
Main Building Exclusive	Restrict reservations in this building to only users for whom it is their assigned main building. Admins are always exempt.	Off

1. Click **Save**.

7.2 Auto No-Show Behavior

When enabled, a background task runs every 5 minutes that:

1. Marks **pending** reservations as **no-show** if the date has passed or the end time has elapsed.
2. Records a **no_show_at timestamp** on the reservation (the moment it was tagged as no-show).
3. Sends a notification to the user about the no-show.
4. Increments the user's no-show count.
5. After the configured release minutes **from the no_show_at timestamp** (not from the reservation end time), releases the slot back to "available" status.

7.3 Waitlist Behavior

When waitlist is enabled for a building:

1. Users who find all slots booked can **join a waitlist** for a specific building and date.
2. When a slot becomes available (due to cancellation or no-show release), the system automatically **notifies the next person** on the waitlist via an in-app notification.
3. The notified user has a **configurable time window** (default 15 minutes) to book a slot.
4. If they don't book within the window, a background task marks their waitlist entry as **expired** and notifies the **next user** in line.
5. Admins can view the full waitlist for any building via the admin waitlist management panel.

8. Attendant Management

8.1 Managing Attendants

Navigate to **Attendants** to view and manage parking attendant accounts. Attendants can be assigned to specific buildings to limit their view.

8.2 Assigning Buildings to Attendants

When creating or editing an attendant user:

1. Click **Add Attendant** or the edit button on an existing attendant.
2. Fill in the required fields (name, email, password).
3. In the **Assigned Buildings** section, check one or more buildings from the list.

4. At least one building **must** be selected — the Create/Save button is disabled otherwise.
5. Selected buildings appear as badges below the checklist.
6. Click **Create** or **Update** to save.

***Note:** Attendants can now be assigned to **multiple buildings** simultaneously. Use the checkbox list to select all buildings the attendant should have access to.*

9. Reservations (Admin View)

9.1 Viewing All Reservations

Navigate to **Reservations** to see all reservations across all buildings.

9.2 Filtering

Filter by:

- **Building**
- **Date**
- **Status** (All, Pending, Confirmed, Cancelled, No Show)
- **Search** (user name, email)

9.3 Admin Cancellation

Click **Cancel** on any active reservation to cancel it on behalf of a user.

1. A **Cancellation Reason** dialog will appear. You **must** provide a reason before confirming.
 2. Click **Confirm Cancellation**.
 3. The reservation status changes to **Cancelled**.
 4. The parker automatically receives an **in-app notification** containing your cancellation reason.
 5. If waitlist is enabled for the building, the next person on the waitlist is notified that a slot has become available.
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10. Reports & Analytics

10.1 Dashboard Stats

The Admin Dashboard shows:

- **Total Reservations, Confirmed, Cancelled, Pending, No-Shows**
- **Buildings Count, Total Slots, Active Users**
- **Occupancy Rate**
- **Daily Breakdown Chart** — Reservations over time
- **Building Breakdown** — Reservations per building

10.2 Filtering Reports

On the Reports page:

1. Set a **date range** (Start Date / End Date).
2. Optionally filter by **Building**.
3. Stats update automatically.

10.3 AI-Powered Insights

1. On the Reports page, click "**Generate AI Insight**".
2. Optionally select a building for building-specific analysis.
3. The system sends parking data to an AI model and returns **3-5 actionable business insights**.
4. Insights are saved to history and displayed in markdown format.

Note: AI Insights call out to a hosted LLM. The feature is governed by the `AI_INSIGHTS_ENABLED` runtime flag (§10.4).

10.4 Enable / Disable AI Insights

AI Insights is the **only** feature that requires public-internet egress. Ops can flip it without rebuilding images:

Posture	Server config	UI behaviour
Force on	<code>AI_INSIGHTS_ENABLED=true</code>	"AI Insights" button visible · button works if LLM key is set, else returns a clear error at click time
Force off	<code>AI_INSIGHTS_ENABLED=false</code>	Button hidden on Dashboard · Insights card hidden on Reports · API returns <code>503 disabled</code>
	unset	

Posture	Server config	UI behaviour
Auto (default)		Auto-enabled iff <code>EMERGENT_LLM_KEY</code> is non-empty

The frontend reads the live state from `GET /api/system/features` on tab load — no rebuild required when Ops flips the flag.

10.5 Insights History

Click the **"History"** tab on Reports to view previously generated insights. You can delete old insights.

11. Site Customization (Settings)

11.1 Login Page Branding

Navigate to **Settings** to customize the login page:

- **Heading Line 1** — First line of the heading
- **Heading Highlight** — Highlighted (colored) word
- **Heading Line 3** — Third line of the heading
- **Description** — Subtitle text
- **Badge 1 Text** — First feature badge
- **Badge 2 Text** — Second feature badge
- **Announcement** — Optional announcement banner

Click **Save** to apply changes. The login page updates immediately.

12. Building Policies

Building policies allow admins to configure special parking rules for specific buildings (e.g., reserved spots for designated users and vehicles).

12.1 Policy Management

Navigate to **Building Policies** (or the building-specific policy page) to configure:

- **User Assignment** — Select users who are allowed to book specific slots. The user dropdown is **filtered by the building's zone**, showing only users assigned to the relevant zone.
- **Vehicle Selection** — Instead of typing a vehicle registration manually, use the **vehicle registration dropdown** to select from the user's registered vehicles.

12.2 Zone-Filtered User Selection

When assigning users to a building policy:

- The system automatically filters the user list to show only users who belong to the **same zone** as the building.
- This prevents assigning users from unrelated zones to building-specific policies.

12.3 Vehicle-Level Sticker Enforcement

For buildings with `requires_sticker` enabled:

- During booking, the system validates the **specific vehicle** selected — not just the user. The vehicle's plate number must have an active slot registration in the building.
- If a user has multiple vehicles but only one is registered, they can only book with the registered vehicle.
- Error message example: *"Vehicle ABC1234 does not have a parking sticker/registration for this building."*

12.4 Slot-Level Vehicle Enforcement

For non-open floors in policy buildings:

- The system validates that the **specific vehicle** is registered to the **specific slot** being booked.
- A user may be registered to slot B1-03 with vehicle ABC1234 — they can only book B1-03 with that vehicle, not B1-01.
- Error message example: *"Vehicle ABC1234 is not registered to this parking slot."*

12.5 Open Floors

Floors listed in the policy's `open_floor_ids` bypass the strict slot-assignment check. Any user with a valid building registration can book any available slot on an open floor.

12.6 Plate Number Uniqueness

The system enforces **global plate number uniqueness**: no two vehicles across any users can share the same plate number. If an admin or user tries to register a vehicle with a plate already in the system, it is rejected with a clear error message.

13. Event Blocking

The **Event Blocking** module allows you to reserve multiple parking slots for corporate events, preventing regular users from booking those slots during the event period.

13.1 Creating an Event Block

1. Navigate to **Event Blocking** in the admin sidebar.
2. Select a **Building** and **Floor**.
3. Set the **Event Date**, **Start Time**, and **End Time** for the event.
4. The **slot map** will appear, showing available (green) and already-booked (red) slots. Click one or more available slots to select them.
5. Enter an **Event Name** (e.g., "Board Meeting") and a **Reason** (e.g., "Reserved for executive parking").
6. Click **Block Slots**.

The selected slots are immediately blocked — they will appear as unavailable in the regular booking interface for that date and time.

13.2 Viewing Event Blocks

The page displays a list of all active event blocks with:

- Event name and reason
- Building, floor, and slot count
- Date and time range
- Number of slots blocked

13.3 Deleting an Event Block

1. Find the event block in the list.
2. Click the **Delete** button.
3. Confirm deletion.

All reservation records associated with the event block are removed, freeing the slots back to regular users.

Note: Deleting an event block does **not** cancel individual user reservations. It only removes the event-block-type reservations that were created specifically for the event.

14. Database & System Health (multi-DB awareness)

The platform supports two database backends interchangeably; the active one is selected at deploy time. Every admin page shows a small **DB badge** in the sidebar:

- **Green / "MongoDB"** — local or in-cluster MongoDB. Typical for staging.

- **Red / "Couchbase Capella" or "Couchbase Enterprise"** — managed Capella cloud or self-hosted on-prem cluster. Typical for production / HCS.

14.1 Health card

The **Database Health Card** on the Dashboard shows live connection status, host, bucket/database name, and per-collection document counts. Use it to:

- Verify a fresh deployment connected to the right cluster.
- Spot a misrouted environment (e.g. UAT pod accidentally pointing at prod).
- Check post-migration row counts after a Mongo → Couchbase mirror.

14.2 Switching backend

In production this is an **infrastructure change** (edit ConfigMap → restart pods), not an in-app toggle — see the deployment runbook. The app itself simply reads `DB_TYPE` at startup.

***Floor-layout images** are stored separately (Huawei OBS or local volume, depending on deployment). They survive pod restarts and replica scaling. No admin action needed.*

15. Troubleshooting

Issue	Solution
Cannot create user (email exists)	Email must be unique across the system. Check for existing users with the same email.
AI Insights button missing	The deployment has <code>AI_INSIGHTS_ENABLED=false</code> (see §10.4) or <code>EMERGENT_LLM_KEY</code> is empty in auto-mode. Ask Ops to flip the flag if business approves the LLM call.
Floor-layout image uploads correctly but doesn't appear after pod restart	In OBS-backed deployments, image is fetched via <code>/api/uploads/...</code> . Hard-refresh the browser. If 404 persists, the OBS bucket / credentials may be misconfigured — check pod logs.
Database badge says "MongoDB" but you expect Couchbase	Wrong <code>DB_TYPE</code> env var on the pod, or wrong ConfigMap applied. Compare with the deployment runbook.
Bulk upload skipping users	Users with emails already in the system are automatically skipped. Check the summary report.
Cannot block slot (active reservations)	Cancel active reservations for the slot before setting it to maintenance.

Issue	Solution
AI insights not working	Ensure the EMERGENT_LLM_KEY is configured and has sufficient balance.
Zone changes not reflected	Users may need to refresh their browser to see updated zone assignments.
Waitlist not available	Ensure waitlist is enabled in the building's Parking Config settings.
Duplicate plate number error	The plate is already registered to another vehicle. Check the existing vehicles list to resolve.
"Vehicle does not have sticker" error	The vehicle's plate must have an active slot registration in the building. Go to Building Policies to add the registration.
"Vehicle not registered to slot" error	For non-open floors, the vehicle must be assigned to the specific slot. Update the slot registration with the correct vehicle.
Users cannot book in exclusive building	Ensure their main_building is set correctly. Or disable Main Building Exclusive in Parking Config.
Event block slot not visible	Ensure the correct building, floor, and date are selected. Already-booked slots are shown in red and cannot be blocked.
Event block delete not releasing slots	Only event-block-type reservations are removed. Regular user reservations are not affected.

Related: [End User Guide](#) | [Attendant Guide](#)